

Control flow and loops

Module 1 - Lesson 3

Overview

Control flow lets programs make decisions and repeat work. if/elif/else branches and for/while loops are the basic grammar of every Python program.

Key Concepts

- if, elif, and else evaluate conditions in order and run the first true branch.
- Boolean operators and, or, not combine conditions; in tests membership.
- for loops iterate over any iterable: lists, strings, dicts, ranges.
- range(n) produces the numbers 0..n-1; enumerate() adds indexes.
- while loops repeat until a condition is false; guard against infinite loops.
- break exits a loop early; continue skips to the next iteration.

Example

```
for i, fruit in enumerate(["apple", "banana", "cherry"]):
    if i == 1:
        continue
    print(i, fruit)

scores = [85, 92, 47]
grade = "Pass" if all(s >= 50 for s in scores) else "Fail"
```

Summary

Conditionals route execution and loops repeat it. Combined with range, enumerate, and break/continue, they express nearly any iteration logic.

Practice Checklist

- Print the first 10 even numbers with a for loop.
- Rewrite a while loop as a for loop and compare readability.
- Write a function that returns the max value without using max().